

Term 3 Practice Activity: C4 Variance and Standard Deviation

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Evaluated criteria

C4 Interprets the concept of variance and standard deviation of a probability function.

1 Uniform Distribution

Problem. A student's daily streaming time X (minutes) is modeled by

$$X \sim \text{Uniform}(2, 14).$$

Questions/Tasks.

- a) Find $E[X]$.
- b) Find $\text{Var}(X)$ and the standard deviation.
- c) Write the PDF of X .
- d) Draw the graph of the PDF, indicate $E[X]$, and draw an interval of one standard deviation around $E[X]$.
- e) Calculate the probability of the highlighted interval.
- f) Compare the probabilities of intervals of the same length around 4 and around 12. What is the difference between these probabilities?

C4

For $X \sim \text{Uniform}(a, b)$:

$$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

- a) With $a = 2$ and $b = 14$,

$$E[X] = \frac{2+14}{2} = 8.$$

- b) Using the same endpoints,

$$\text{Var}(X) = \frac{(14-2)^2}{12} = \frac{144}{12} = 12,$$

so

$$\sigma = \sqrt{12} = 2\sqrt{3} \approx 3.46.$$

- c) Since $X \sim \text{Uniform}(2, 14)$, the PDF is constant on the support $[2, 14]$. Its height is

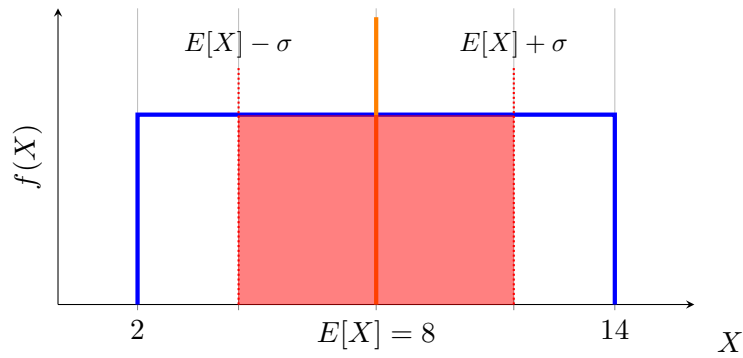
$$\frac{1}{14-2} = \frac{1}{12}.$$

Therefore,

$$f(x) = \begin{cases} \frac{1}{12}, & 2 \leq x \leq 14, \\ 0, & \text{otherwise.} \end{cases}$$

- d) From parts a and b, the mean is $E[X] = 8$ and the standard deviation is $\sigma = 2\sqrt{3}$. Therefore, one standard deviation around the mean is

$$[E[X] - \sigma, E[X] + \sigma] = [8 - 2\sqrt{3}, 8 + 2\sqrt{3}] \approx [4.54, 11.46].$$



e) The highlighted interval is the interval found in part d:

$$\left[8 - 2\sqrt{3}, 8 + 2\sqrt{3}\right].$$

So the probability of the highlighted interval is

$$P\left(8 - 2\sqrt{3} \leq X \leq 8 + 2\sqrt{3}\right).$$

Using the PDF from part c, the height of the rectangle is $\frac{1}{12}$. The base of the rectangle is the length of the highlighted interval:

$$\left(8 + 2\sqrt{3}\right) - \left(8 - 2\sqrt{3}\right) = 4\sqrt{3}.$$

Therefore,

$$P\left(8 - 2\sqrt{3} \leq X \leq 8 + 2\sqrt{3}\right) = (\text{base})(\text{height}) = (4\sqrt{3})\left(\frac{1}{12}\right).$$

Thus,

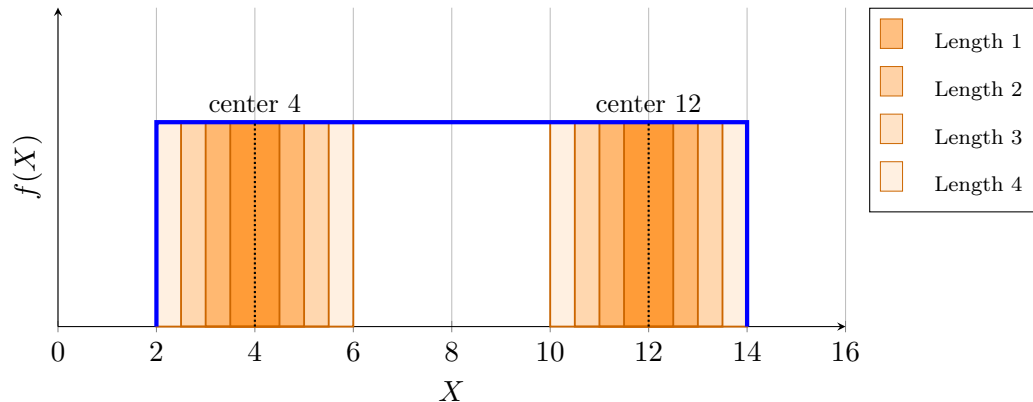
$$P\left(8 - 2\sqrt{3} \leq X \leq 8 + 2\sqrt{3}\right) = \frac{4\sqrt{3}}{12} = \frac{\sqrt{3}}{3} \approx 0.577.$$

f) Since this is a continuous model, the probability of any exact single value is 0. So we compare intervals of the same length around 4 and around 12.

Using the PDF from part c, any interval inside $[2, 14]$ with length L has probability

$$P(\text{interval}) = L\left(\frac{1}{12}\right) = \frac{L}{12}.$$

The following graph shows the 8 intervals. Intervals of length 1 are the darkest, and intervals of length 4 are the lightest.



Interval length	Around 4	Probability	Around 12	Probability	Difference
1	[3.5, 4.5]	$\frac{1}{12}$	[11.5, 12.5]	$\frac{1}{12}$	$\frac{0}{12}$
2	[3, 5]	$\frac{2}{12}$	[11, 13]	$\frac{2}{12}$	$\frac{0}{12}$
3	[2.5, 5.5]	$\frac{3}{12}$	[10.5, 13.5]	$\frac{3}{12}$	$\frac{0}{12}$
4	[2, 6]	$\frac{4}{12}$	[10, 14]	$\frac{4}{12}$	$\frac{0}{12}$

In each row, the two intervals have the same length and lie completely inside the support $[2, 14]$. Since the distribution is uniform, their probabilities are equal.

The difference between the probabilities around 4 and around 12 is always

0.

2 Triangular Distribution 1

Problem. A bus wait time W (minutes) is modeled by a triangular distribution with minimum $a = 2$, mode $c = 2$, and maximum $b = 14$:

$$W \sim \text{Triangular}(2, 2, 14).$$

Questions/Tasks.

- Find $E[W]$.
- Find $\text{Var}(W)$ and the standard deviation.
- Write the PDF of W .
- Draw the graph of the PDF, indicate $E[W]$, and draw an interval of one standard deviation around $E[W]$.
- Calculate the probability of the highlighted interval.
- Compare the probabilities of intervals of the same length around 4 and around 12. What is the difference between these probabilities?

C4

For $\text{Triangular}(a, c, b)$:

$$E[W] = \frac{a + b + c}{3}, \quad \text{Var}(W) = \frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}.$$

- a) With $a = 2$, $c = 2$, and $b = 14$,

$$E[W] = \frac{2 + 14 + 2}{3} = \frac{18}{3} = 6.$$

- b) Using the same parameters,

$$\text{Var}(W) = \frac{2^2 + 14^2 + 2^2 - (2)(14) - (2)(2) - (14)(2)}{18} = \frac{4 + 196 + 4 - 28 - 4 - 28}{18} = \frac{144}{18} = 8.$$

Therefore,

$$\sigma = \sqrt{8} = 2\sqrt{2} \approx 2.83.$$

- c) Since the mode is at the left endpoint, the PDF is a decreasing straight line from $w = 2$ to $w = 14$. The total area under the PDF must be 1. Therefore, the maximum height is found from the area of the full triangle:

$$A_{\triangle} = \frac{1}{2}(\text{base})(\text{height}) = 1 \quad , \quad \text{base} = 14 - 2 = 12.$$

So

$$A_{\triangle} = \frac{1}{2}(12)(\text{height}) = 1 \quad \longrightarrow \quad 6(\text{height}) = 1 \quad \longrightarrow \quad \text{height} = \frac{1}{6}.$$

The PDF has height $\frac{1}{6}$ at $w = 2$ and height 0 at $w = 14$. Since this is one decreasing line segment, use the basic line form

$$f(w) = m(w - c),$$

To find c , use the fact that the line has height 0 at $w = 14$:

$$f(14) = 0 \quad \longrightarrow \quad m(14 - c) = 0 \quad \longrightarrow \quad 14 - c = 0 \quad \longrightarrow \quad c = 14.$$

Therefore, the line has the form

$$f(w) = m(w - 14), \quad 2 \leq w \leq 14.$$

To find m , use the fact that the line has height $\frac{1}{6}$ at the mode $w = 2$:

$$f(2) = \frac{1}{6} \quad \longrightarrow \quad m(2 - 14) = \frac{1}{6} \quad \longrightarrow \quad -12m = \frac{1}{6} \quad \longrightarrow \quad m = -\frac{1}{72}.$$

Therefore,

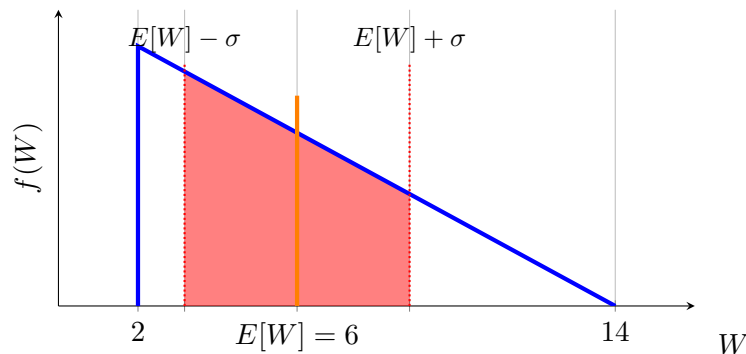
$$f(w) = -\frac{1}{72}(w - 14) = \frac{14 - w}{72}, \quad 2 \leq w \leq 14.$$

Thus,

$$f(w) = \begin{cases} \frac{14-w}{72}, & 2 \leq w \leq 14, \\ 0, & \text{otherwise.} \end{cases}$$

- d) From parts a and b, the mean is $E[W] = 6$ and the standard deviation is $\sigma = 2\sqrt{2}$. Therefore, one standard deviation around the mean is

$$[E[W] - \sigma, E[W] + \sigma] = [6 - 2\sqrt{2}, 6 + 2\sqrt{2}] \approx [3.17, 8.83].$$



- e) The highlighted interval is the interval found in part d:

$$[6 - 2\sqrt{2}, 6 + 2\sqrt{2}].$$

The probability of this interval is represented by the area under the PDF above the highlighted interval:

$$P(6 - 2\sqrt{2} \leq W \leq 6 + 2\sqrt{2}).$$

Since the PDF is a straight line over this interval, the highlighted region is a trapezoid. Therefore,

$$P(6 - 2\sqrt{2} \leq W \leq 6 + 2\sqrt{2}) = \frac{1}{2}(\text{width})(\text{left height} + \text{right height}).$$

Using the PDF from part c, the height at the left endpoint of the highlighted interval is

$$\text{left height} = f(6 - 2\sqrt{2}) = \frac{14 - (6 - 2\sqrt{2})}{72} = \frac{8 + 2\sqrt{2}}{72} = \frac{4 + \sqrt{2}}{36}.$$

The height at the right endpoint is

$$\text{right height} = f(6 + 2\sqrt{2}) = \frac{14 - (6 + 2\sqrt{2})}{72} = \frac{8 - 2\sqrt{2}}{72} = \frac{4 - \sqrt{2}}{36}.$$

The width of the trapezoid is

$$\text{width} = (6 + 2\sqrt{2}) - (6 - 2\sqrt{2}) = 4\sqrt{2}.$$

Substituting the width and the two heights into the trapezoid area formula,

$$P(6 - 2\sqrt{2} \leq W \leq 6 + 2\sqrt{2}) = \frac{1}{2}(4\sqrt{2}) \left(\frac{4 + \sqrt{2}}{36} + \frac{4 - \sqrt{2}}{36} \right).$$

Since

$$\frac{4 + \sqrt{2}}{36} + \frac{4 - \sqrt{2}}{36} = \frac{8}{36} = \frac{2}{9},$$

we get

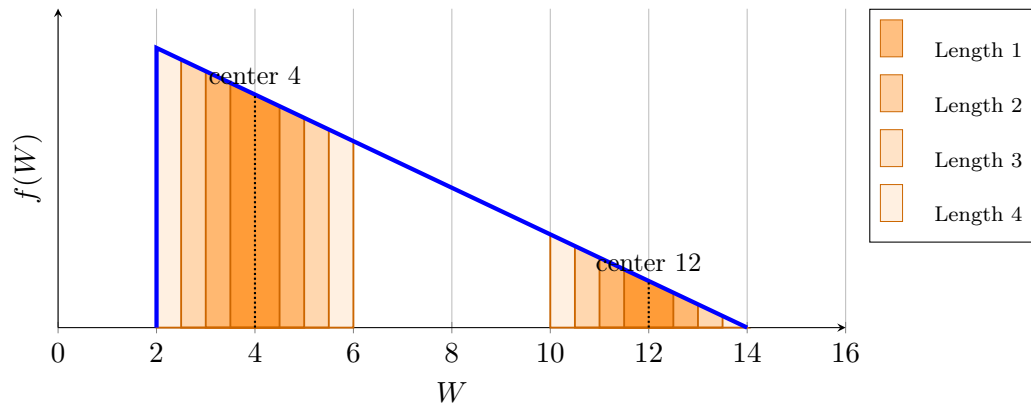
$$P(6 - 2\sqrt{2} \leq W \leq 6 + 2\sqrt{2}) = \frac{1}{2}(4\sqrt{2}) \left(\frac{2}{9} \right) = \frac{4\sqrt{2}}{9} \approx 0.629.$$

- f) Since this is a continuous model, the probability of any exact single value is 0. So we compare intervals of the same length around 4 and around 12.

Using the PDF from part c, each interval probability is found as the area of a trapezoid:

$$P(r \leq W \leq s) = \frac{1}{2}(\text{interval length})(f(r) + f(s)).$$

The following graph shows the 8 intervals. Intervals of length 1 are the darkest, and intervals of length 4 are the lightest.



Interval length	Around 4	Probability	Around 12	Probability	Difference
1	[3.5, 4.5]	$\frac{5}{36}$	[11.5, 12.5]	$\frac{1}{36}$	$\frac{4}{36}$
2	[3, 5]	$\frac{10}{36}$	[11, 13]	$\frac{2}{36}$	$\frac{8}{36}$
3	[2.5, 5.5]	$\frac{15}{36}$	[10.5, 13.5]	$\frac{3}{36}$	$\frac{12}{36}$
4	[2, 6]	$\frac{20}{36}$	[10, 14]	$\frac{4}{36}$	$\frac{16}{36}$

The intervals around 4 have larger probability than the intervals around 12. This happens because the density is higher near the left side of the distribution and decreases as W increases.

3 Triangular Distribution 2

Problem. A bus wait time W (minutes) is modeled by a triangular distribution with minimum $a = 2$, mode $c = 6$, and maximum $b = 14$:

$$W \sim \text{Triangular}(2, 6, 14).$$

Questions/Tasks.

- Find $E[W]$.
- Find $\text{Var}(W)$ and the standard deviation.
- Write the PDF of W .
- Draw the graph of the PDF, indicate $E[W]$, and draw an interval of one standard deviation around $E[W]$.
- Calculate the probability of the highlighted interval.
- Compare the probabilities of intervals of the same length around 4 and around 12. What is the difference between these probabilities?

C4

For $\text{Triangular}(a, c, b)$:

$$E[W] = \frac{a + b + c}{3}, \quad \text{Var}(W) = \frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}.$$

- a) With $a = 2$, $c = 6$, and $b = 14$,

$$E[W] = \frac{2 + 14 + 6}{3} = \frac{22}{3} \approx 7.33.$$

- b) Using the same parameters,

$$\text{Var}(W) = \frac{2^2 + 14^2 + 6^2 - (2)(14) - (2)(6) - (14)(6)}{18} = \frac{4 + 196 + 36 - 28 - 12 - 84}{18} = \frac{112}{18} = \frac{56}{9}.$$

Therefore,

$$\sigma = \sqrt{\frac{56}{9}} = \frac{2\sqrt{14}}{3} \approx 2.49.$$

- c) Since the mode is at $w = 6$, the PDF increases linearly from $w = 2$ to $w = 6$, then decreases linearly from $w = 6$ to $w = 14$. The total area under the PDF must be 1. Therefore, the maximum height is found from the area of the full triangle:

$$A_{\Delta} = \frac{1}{2}(\text{base})(\text{height}) = 1, \quad \text{base} = 14 - 2 = 12.$$

So

$$A_{\Delta} = \frac{1}{2}(12)(\text{height}) = 1 \quad \longrightarrow \quad 6(\text{height}) = 1 \quad \longrightarrow \quad \text{height} = \frac{1}{6}.$$

The graph has two line segments. For each segment, use the basic line form

$$f(w) = m(w - c).$$

For the left side, the line has height 0 at $w = 2$. To find c , use

$$f(2) = 0 \quad \longrightarrow \quad m_1(2 - c) = 0 \quad \longrightarrow \quad 2 - c = 0 \quad \longrightarrow \quad c = 2.$$

Therefore, the left side has the form

$$f(w) = m_1(w - 2), \quad 2 \leq w \leq 6.$$

To find m_1 , use the fact that the line has height $\frac{1}{6}$ at the mode $w = 6$:

$$f(6) = \frac{1}{6} \quad \longrightarrow \quad m_1(6 - 2) = \frac{1}{6} \quad \longrightarrow \quad 4m_1 = \frac{1}{6} \quad \longrightarrow \quad m_1 = \frac{1}{24}.$$

Therefore,

$$f(w) = \frac{1}{24}(w - 2) = \frac{w - 2}{24}, \quad 2 \leq w \leq 6.$$

For the right side, the line has height 0 at $w = 14$. To find c , use

$$f(14) = 0 \quad \longrightarrow \quad m_2(14 - c) = 0 \quad \longrightarrow \quad 14 - c = 0 \quad \longrightarrow \quad c = 14.$$

Therefore, the right side has the form

$$f(w) = m_2(w - 14), \quad 6 \leq w \leq 14.$$

To find m_2 , use the fact that the line has height $\frac{1}{6}$ at the mode $w = 6$:

$$f(6) = \frac{1}{6} \quad \longrightarrow \quad m_2(6 - 14) = \frac{1}{6} \quad \longrightarrow \quad -8m_2 = \frac{1}{6} \quad \longrightarrow \quad m_2 = -\frac{1}{48}.$$

Therefore,

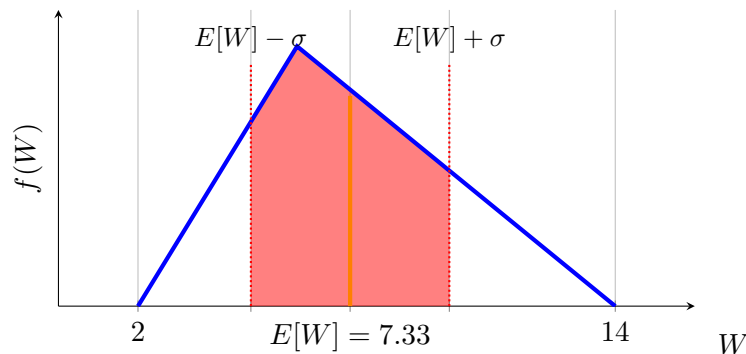
$$f(w) = -\frac{1}{48}(w - 14) = \frac{14 - w}{48}, \quad 6 \leq w \leq 14.$$

Thus,

$$f(w) = \begin{cases} \frac{w-2}{24}, & 2 \leq w \leq 6, \\ \frac{14-w}{48}, & 6 \leq w \leq 14, \\ 0, & \text{otherwise.} \end{cases}$$

- d) From parts a and b, the mean is $E[W] = \frac{22}{3}$ and the standard deviation is $\sigma = \frac{2\sqrt{14}}{3}$. Therefore, one standard deviation around the mean is

$$[E[W] - \sigma, E[W] + \sigma] = \left[\frac{22 - 2\sqrt{14}}{3}, \frac{22 + 2\sqrt{14}}{3} \right] \approx [4.84, 9.83].$$



e) The highlighted interval is the interval found in part d:

$$\left[\frac{22 - 2\sqrt{14}}{3}, \frac{22 + 2\sqrt{14}}{3} \right].$$

The probability of this interval is represented by the area under the PDF above the highlighted interval:

$$P\left(\frac{22 - 2\sqrt{14}}{3} \leq W \leq \frac{22 + 2\sqrt{14}}{3}\right).$$

Instead of calculating the highlighted area directly, use the complement rule. The total area under the PDF is 1, so the highlighted probability is

$$1 - (\text{left extreme triangle}) - (\text{right extreme triangle}).$$

Using the PDF from part c, the height at the left endpoint of the highlighted interval is

$$\text{left height} = f\left(\frac{22 - 2\sqrt{14}}{3}\right) = \frac{\frac{22-2\sqrt{14}}{3} - 2}{24} = \frac{8 - \sqrt{14}}{36}.$$

The base of the left extreme triangle is

$$\text{left base} = \frac{22 - 2\sqrt{14}}{3} - 2 = \frac{16 - 2\sqrt{14}}{3}.$$

Therefore, the area of the left extreme triangle is

$$A_{L\Delta} = \frac{1}{2} \left(\frac{16 - 2\sqrt{14}}{3} \right) \left(\frac{8 - \sqrt{14}}{36} \right) = \frac{13}{18} - \frac{4\sqrt{14}}{27}.$$

Using the PDF from part c, the height at the right endpoint of the highlighted interval is

$$\text{right height} = f\left(\frac{22 + 2\sqrt{14}}{3}\right) = \frac{14 - \frac{22+2\sqrt{14}}{3}}{48} = \frac{10 - \sqrt{14}}{72}.$$

The base of the right extreme triangle is

$$\text{right base} = 14 - \frac{22 + 2\sqrt{14}}{3} = \frac{20 - 2\sqrt{14}}{3}.$$

Therefore, the area of the right extreme triangle is

$$A_{R\Delta} = \frac{1}{2} \left(\frac{20 - 2\sqrt{14}}{3} \right) \left(\frac{10 - \sqrt{14}}{72} \right) = \frac{19}{36} - \frac{5\sqrt{14}}{54}.$$

By the complement rule,

$$P\left(\frac{22 - 2\sqrt{14}}{3} \leq W \leq \frac{22 + 2\sqrt{14}}{3}\right) = 1 - A_{L\Delta} - A_{R\Delta}$$

Substituting the two triangle areas,

$$P\left(\frac{22 - 2\sqrt{14}}{3} \leq W \leq \frac{22 + 2\sqrt{14}}{3}\right) = 1 - \left(\frac{13}{18} - \frac{4\sqrt{14}}{27}\right) - \left(\frac{19}{36} - \frac{5\sqrt{14}}{54}\right).$$

Thus,

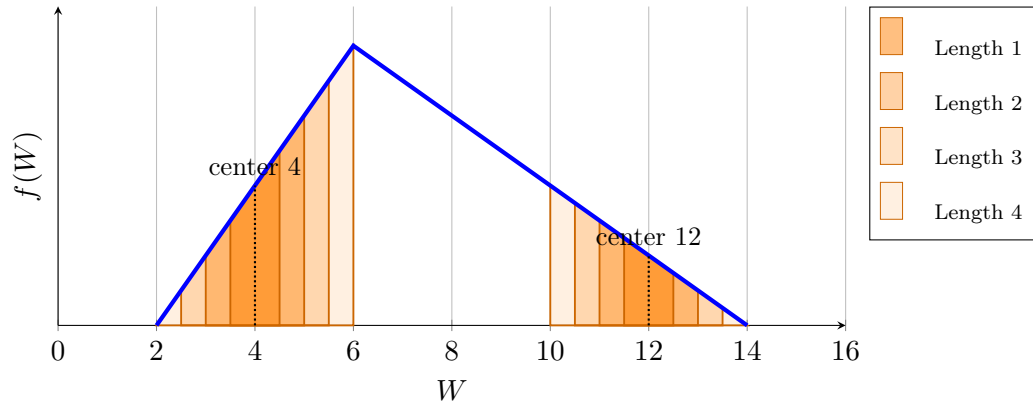
$$P\left(\frac{22 - 2\sqrt{14}}{3} \leq W \leq \frac{22 + 2\sqrt{14}}{3}\right) = -\frac{1}{4} + \frac{13\sqrt{14}}{54} \approx 0.651.$$

- f) Since this is a continuous model, the probability of any exact single value is 0. So we compare intervals of the same length around 4 and around 12.

Using the PDF from part c, each interval probability is found as the area of a trapezoid:

$$P(r \leq W \leq s) = \frac{1}{2}(\text{interval length})(f(r) + f(s)).$$

The following graph shows the 8 intervals. Intervals of length 1 are the darkest, and intervals of length 4 are the lightest.



Interval length	Around 4	Probability	Around 12	Probability	Difference
1	[3.5, 4.5]	$\frac{2}{24}$	[11.5, 12.5]	$\frac{1}{24}$	$\frac{1}{24}$
2	[3, 5]	$\frac{4}{24}$	[11, 13]	$\frac{2}{24}$	$\frac{2}{24}$
3	[2.5, 5.5]	$\frac{6}{24}$	[10.5, 13.5]	$\frac{3}{24}$	$\frac{3}{24}$
4	[2, 6]	$\frac{8}{24}$	[10, 14]	$\frac{4}{24}$	$\frac{4}{24}$

The intervals around 4 have larger probability than the intervals around 12. This happens because 4 is closer to the mode than 12, so the density is higher around 4.

4 Normal Distribution

Problem. Quiz scores S are modeled by

$$S \sim N(6, 4^2).$$

Questions/Tasks.

- State $E[S]$.
- State $\text{Var}(S)$ and the standard deviation.
- Write the PDF of S .
- Give the intervals $[\mu - \sigma, \mu + \sigma]$ and $[\mu - 2\sigma, \mu + 2\sigma]$, draw the graph of the PDF, indicate $E[S]$, and draw the interval of one standard deviation around $E[S]$.
- Find the probability of the highlighted interval $[\mu - \sigma, \mu + \sigma]$.
- Compare the probabilities of intervals of the same length around 6 and around -2 . What is the difference between these probabilities?

C4

For $N(\mu, \sigma^2)$,

$$E[S] = \mu, \quad \text{Var}(S) = \sigma^2.$$

a) Since

$$S \sim N(6, 4^2),$$

the mean is

$$E[S] = \mu = 6.$$

b) Using the same parameters,

$$\text{Var}(S) = \sigma^2 = 4^2 = 16.$$

Therefore,

$$\sigma = 4.$$

c) For a normal distribution, the PDF is

$$f(s) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(s-\mu)^2}{2\sigma^2}}.$$

Using $\mu = 6$ and $\sigma = 4$, the PDF is

$$f(s) = \frac{1}{4\sqrt{2\pi}} e^{-\frac{(s-6)^2}{2(4)^2}}.$$

Equivalently,

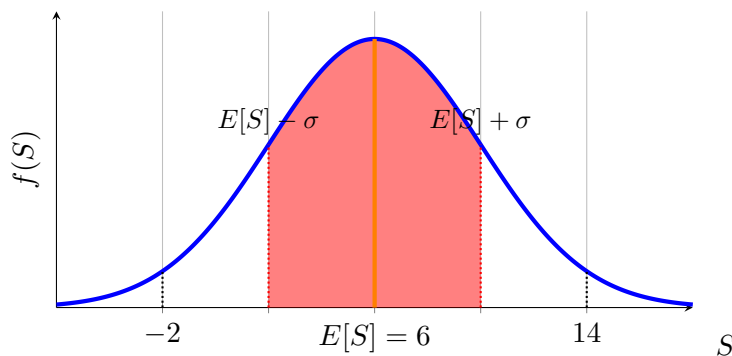
$$f(s) = \frac{1}{4\sqrt{2\pi}} e^{-\frac{(s-6)^2}{32}}, \quad -\infty < s < \infty.$$

d) From parts a and b, the mean is $E[S] = 6$ and the standard deviation is $\sigma = 4$. Therefore,

$$[\mu - \sigma, \mu + \sigma] = [6 - 4, 6 + 4] = [2, 10].$$

Also,

$$[\mu - 2\sigma, \mu + 2\sigma] = [6 - 2(4), 6 + 2(4)] = [-2, 14].$$



e) The highlighted interval is the interval found in part d:

$$[2, 10].$$

The probability of this interval is represented by the area under the normal curve above the highlighted interval:

$$P(2 \leq S \leq 10).$$

To use the standard normal distribution, standardize with

$$Z = \frac{S - \mu}{\sigma} = \frac{S - 6}{4}.$$

Then

$$P(2 \leq S \leq 10) = P\left(\frac{2-6}{4} \leq Z \leq \frac{10-6}{4}\right).$$

So

$$P(2 \leq S \leq 10) = P(-1 \leq Z \leq 1).$$

Using the standard normal distribution,

$$P(-1 \leq Z \leq 1) \approx 0.6827.$$

Therefore,

$$P(2 \leq S \leq 10) \approx 0.6827.$$

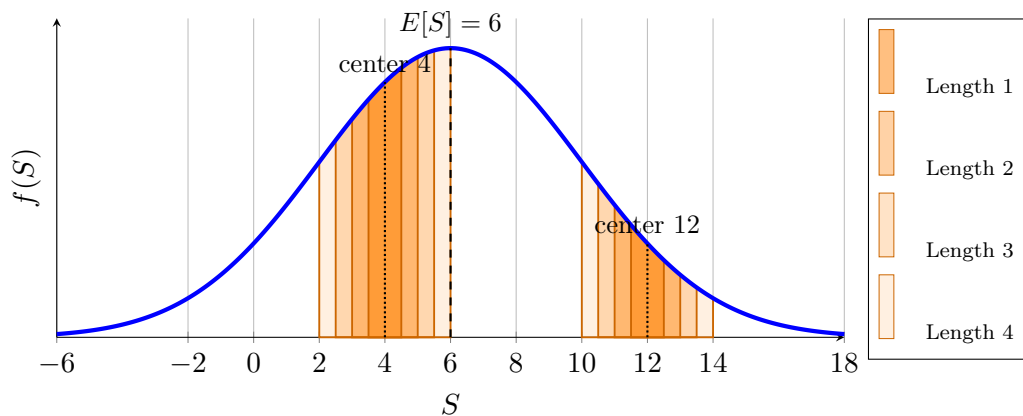
- f) Since this is a continuous model, the probability of any exact single value is 0. So we compare intervals of the same length around 4 and around 12.

Using the standardization formula from part e,

$$Z = \frac{S - 6}{4},$$

each interval probability can be found by converting the endpoints to z -scores and using the standard normal distribution.

The following graph shows the 8 intervals. Intervals of length 1 are the darkest, and intervals of length 4 are the lightest.



Interval length	Around 4	Probability	Around 12	Probability	Difference
1	[3.5, 4.5]	0.0878	[11.5, 12.5]	0.0325	0.0553
2	[3, 5]	0.1747	[11, 13]	0.0656	0.1091
4	[2, 6]	0.3413	[10, 14]	0.1359	0.2054
8	[0, 8]	0.6247	[8, 16]	0.3023	0.3223

The intervals around 4 have larger probability than the intervals around 12. This happens because 4 is closer to the mean 6, where the normal density is highest. The value 12 is farther from the mean, so intervals around 12 have less area under the curve.

5 Comparing Uniform Distributions

Problem. Compare these uniform models for download times (seconds):

$$X_1 \sim \text{Uniform}(4, 10), \quad X_2 \sim \text{Uniform}(6, 12), \quad X_3 \sim \text{Uniform}(4, 14).$$

Questions/Tasks.

- a) Find $E[X_i]$ and $\text{Var}(X_i)$ for $i = 1, 2, 3$.
- b) Compare X_1 and X_2 : only location changes. How do center and spread change?
- c) Compare X_1 and X_3 : only width changes from 6 to 10. How does variance change?
- d) Which model has the widest spread, and why?

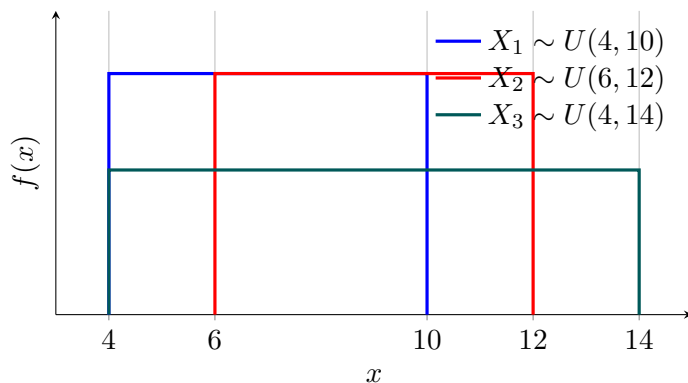
C4

Using $E = \frac{a+b}{2}$, $\text{Var} = \frac{(b-a)^2}{12}$:

$$E[X_1] = 7, \quad \text{Var}(X_1) = \frac{6^2}{12} = 3;$$

$$E[X_2] = 9, \quad \text{Var}(X_2) = \frac{6^2}{12} = 3;$$

$$E[X_3] = 9, \quad \text{Var}(X_3) = \frac{10^2}{12} = \frac{25}{3}.$$



Shifting both endpoints by the same amount changes the mean but not spread. Increasing interval width increases variance.

6 Comparing Triangular Distributions

Problem. A packing process time T (minutes) follows triangular models:

$$T_1 \sim \text{Triangular}(2, 5, 10), \quad T_2 \sim \text{Triangular}(2, 7, 10), \quad T_3 \sim \text{Triangular}(4, 7, 12).$$

Questions/Tasks.

- a) Compute $E[T_i]$ and $\text{Var}(T_i)$ for each model.

- b) Compare T_1 and T_2 : only mode changes. How do center and shape change?
- c) Compare T_2 and T_3 : all three parameters shift by $+2$. What stays the same and what changes?
- d) Which model is most concentrated near larger times? Explain using the peak location.

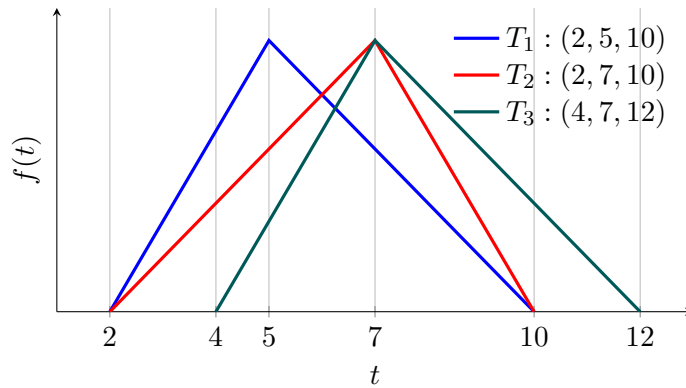
C4

Using triangular formulas:

$$E[T_1] = \frac{2 + 5 + 10}{3} = \frac{17}{3}, \quad \text{Var}(T_1) = \frac{2^2 + 5^2 + 10^2 - 2 \cdot 5 - 2 \cdot 10 - 5 \cdot 10}{18} = \frac{49}{18}.$$

$$E[T_2] = \frac{2 + 7 + 10}{3} = \frac{19}{3}, \quad \text{Var}(T_2) = \frac{2^2 + 7^2 + 10^2 - 2 \cdot 7 - 2 \cdot 10 - 7 \cdot 10}{18} = \frac{49}{18}.$$

$$E[T_3] = \frac{4 + 7 + 12}{3} = \frac{23}{3}, \quad \text{Var}(T_3) = \frac{4^2 + 7^2 + 12^2 - 4 \cdot 7 - 4 \cdot 12 - 7 \cdot 12}{18} = \frac{49}{18}.$$



Changing only the mode shifts peak and mean but here keeps variance unchanged; shifting all parameters by $+2$ shifts location but not spread.

7 Comparing Normal Distributions

Problem. Daily step counts (thousands) are modeled as:

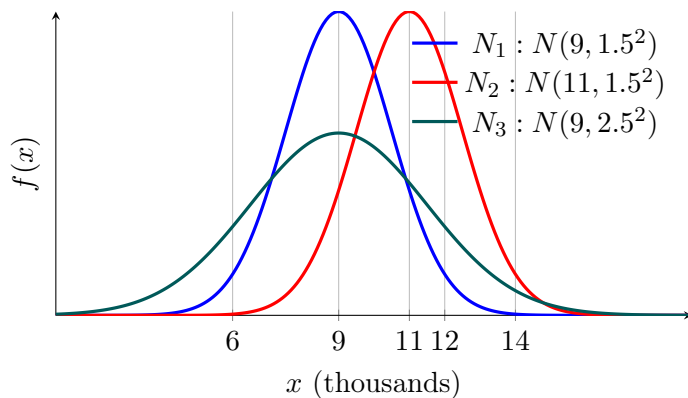
$$N_1 \sim N(9, 1.5^2), \quad N_2 \sim N(11, 1.5^2), \quad N_3 \sim N(9, 2.5^2).$$

Questions/Tasks.

- a) State mean and variance for each distribution.
- b) Compare N_1 and N_2 : only mean changes. What changes on the graph?
- c) Compare N_1 and N_3 : only standard deviation changes. What happens to spread and peak height?
- d) Which distribution gives a higher chance of being far from its mean? Explain.

C4

$$E[N_1] = 9, \text{ Var}(N_1) = 2.25; \quad E[N_2] = 11, \text{ Var}(N_2) = 2.25; \quad E[N_3] = 9, \text{ Var}(N_3) = 6.25.$$



Changing μ moves the center left or right without changing shape width. Increasing σ makes the curve wider and less peaked, so far-from-mean values become more likely.

8 Uniform with Parameters

Problem. A random time X (minutes) is

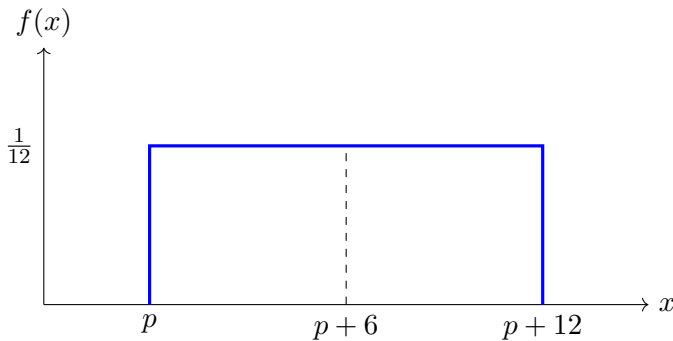
$$X \sim \text{Uniform}(p, p + 12), \quad p \in \mathbb{R}.$$

Questions/Tasks.

- Express $E[X]$ in terms of p .
- Express $\text{Var}(X)$ in terms of p .
- Which parameter controls location and which controls spread?
- If p increases by 5, how do mean and variance change?

C4

$$E[X] = \frac{p + (p + 12)}{2} = p + 6, \quad \text{Var}(X) = \frac{((p + 12) - p)^2}{12} = \frac{12^2}{12} = 12.$$



So p controls location; spread is fixed by width 12 and does not depend on p .

9 Triangular with Parameters

Problem. A project completion time Y (days) is modeled by

$$Y \sim \text{Triangular}(4, 4 + m, 14), \quad 0 \leq m \leq 10.$$

Questions/Tasks.

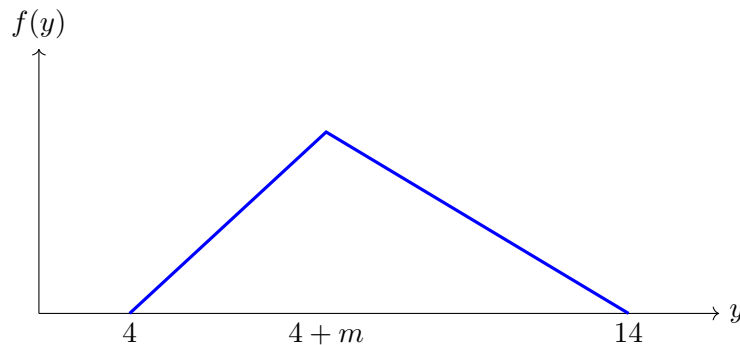
- a) Write $E[Y]$ and $\text{Var}(Y)$ as functions of m .
- b) State the condition on m that makes the model valid.
- c) How does the mean change as m increases?
- d) How does moving the mode affect the shape (left/right skewness)?

C4

Here $a = 4$, $c = 4 + m$, $b = 14$.

$$E[Y] = \frac{a + b + c}{3} = \frac{4 + 14 + (4 + m)}{3} = \frac{22 + m}{3}.$$

$$\text{Var}(Y) = \frac{a^2 + b^2 + c^2 - ab - ac - bc}{18} = \frac{16 + 196 + (4 + m)^2 - 56 - 4(4 + m) - 14(4 + m)}{18} = \frac{m^2 - 10m + 100}{18}.$$



Validity needs $a \leq c \leq b \Rightarrow 4 \leq 4 + m \leq 14 \Rightarrow 0 \leq m \leq 10$. Mean increases linearly with m . As mode moves right, the distribution becomes less right-skewed and can become left-skewed when the mode passes midpoint 9.

10 Normal with Parameters

Problem. A test score variable is

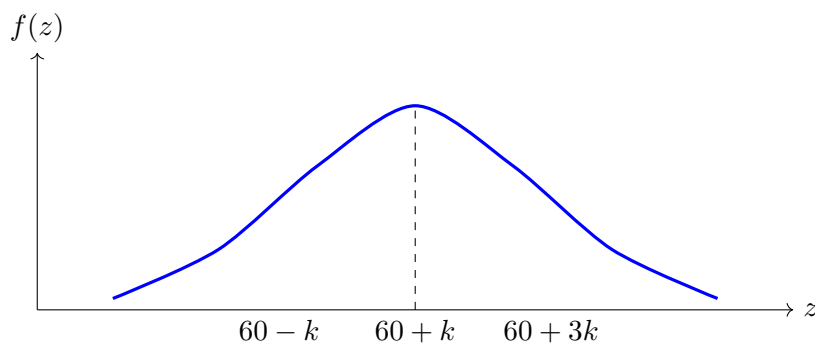
$$Z \sim N(60 + k, (2k)^2), \quad k > 0.$$

Questions/Tasks.

- a) Express mean and variance in terms of k .
- b) Determine valid values of k .
- c) As k increases, how do center and spread change?
- d) Identify which part of the model controls location and which controls spread.

C4

$$E[Z] = 60 + k, \quad \text{Var}(Z) = (2k)^2 = 4k^2, \quad \sigma = 2k.$$



For a valid non-degenerate normal spread, $k > 0$. Increasing k shifts mean right and also increases spread quadratically through variance.

11 Uniform with Parameters

Problem. Commute time C (minutes) is modeled by

$$C \sim \text{Uniform}(10 - r, 10 + r), \quad r > 0.$$

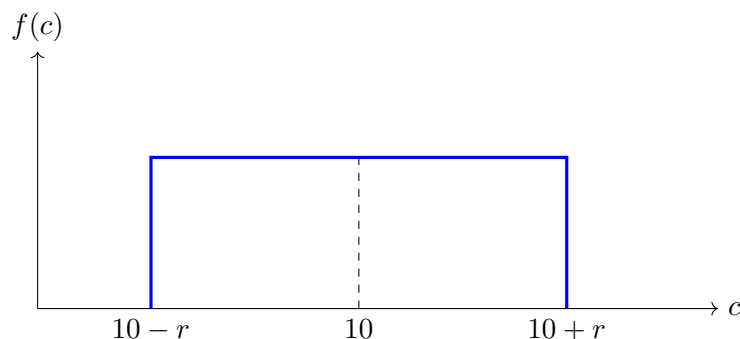
Questions/Tasks.

- Find $E[C]$ and $\text{Var}(C)$ in terms of r .
- What condition on r is needed for a valid interval?
- As r increases, what happens to interval width and spread?
- For what r is the distribution narrower, and why?

C4

$$E[C] = \frac{(10 - r) + (10 + r)}{2} = 10,$$

$$\text{Var}(C) = \frac{((10 + r) - (10 - r))^2}{12} = \frac{(2r)^2}{12} = \frac{r^2}{3}.$$



Need $r > 0$. Width is $2r$, so larger r means wider interval and larger variance; smaller positive r means narrower distribution.

12 Triangular with Parameters

Problem. A completion score Q is modeled by

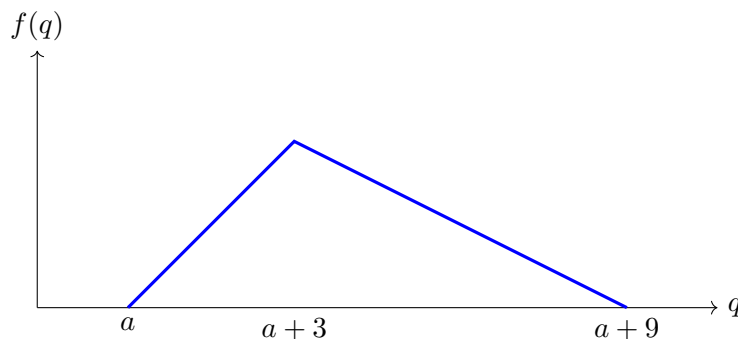
$$Q \sim \text{Triangular}(a, a + 3, a + 9), \quad a \in \mathbb{R}.$$

Questions/Tasks.

- a) Express $E[Q]$ and $\text{Var}(Q)$ in terms of a .
- b) Is this model always valid for any real a ? Explain.
- c) Which parameter controls location? Which controls spread?
- d) If a increases, what happens to center, spread, and shape?

C4

$$E[Q] = \frac{a + (a + 3) + (a + 9)}{3} = a + 4.$$
$$\text{Var}(Q) = \frac{a^2 + (a + 3)^2 + (a + 9)^2 - a(a + 3) - a(a + 9) - (a + 3)(a + 9)}{18} = \frac{7}{2}.$$



Since $a \leq a + 3 \leq a + 9$ always holds, the distribution is valid for all real a . Parameter a shifts location; spread and shape remain unchanged.

13 Normal with Parameters

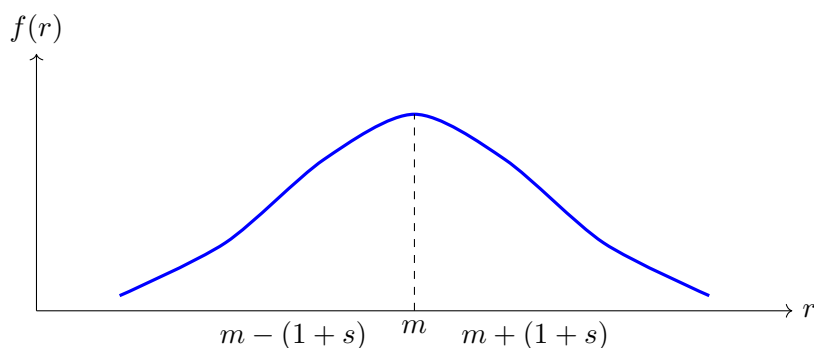
Problem. Reaction time R (seconds) is modeled by

$$R \sim N(m, (1 + s)^2), \quad s > -1.$$

Questions/Tasks.

- a) Express $E[R]$ and $\text{Var}(R)$ in terms of m and s .
- b) Give the condition(s) needed for a valid nonzero spread.
- c) Explain how increasing m changes the distribution.
- d) Explain how increasing s changes the distribution width and likelihood of values far from the mean.

$$E[R] = m, \quad \text{Var}(R) = (1 + s)^2, \quad \sigma = 1 + s.$$



A valid standard deviation needs $\sigma > 0 \Rightarrow 1 + s > 0 \Rightarrow s > -1$. Increasing m shifts the center only. Increasing s increases spread, making the bell wider and less concentrated around the mean.

14 Uniform with Parameters

Problem. A delivery time U (minutes) for a local route is modeled by

$$U \sim \text{Uniform}(h - 2w, h + w), \quad w > 0.$$

Questions/Tasks.

- Derive $E[U]$, $\text{Var}(U)$, and σ_U in terms of h and w .
- Identify which parameter controls the center and which controls the interval width.
- Compare Case A: $(h, w) = (30, 4)$ and Case B: $(h, w) = (35, 4)$. Does the distribution shift, widen, or both?
- Compare Case C: $(h, w) = (30, 4)$ and Case D: $(h, w) = (30, 7)$. How do center, width, and spread change?
- If both h and w increase by 2 at the same time, explain whether the model shifts, widens, or both.

For $U \sim \text{Uniform}(a, b)$, $E[U] = \frac{a+b}{2}$, $\text{Var}(U) = \frac{(b-a)^2}{12}$. Here $a = h - 2w$, $b = h + w$, so

$$E[U] = \frac{(h - 2w) + (h + w)}{2} = h - \frac{w}{2},$$

$$\text{Var}(U) = \frac{((h + w) - (h - 2w))^2}{12} = \frac{(3w)^2}{12} = \frac{3w^2}{4},$$

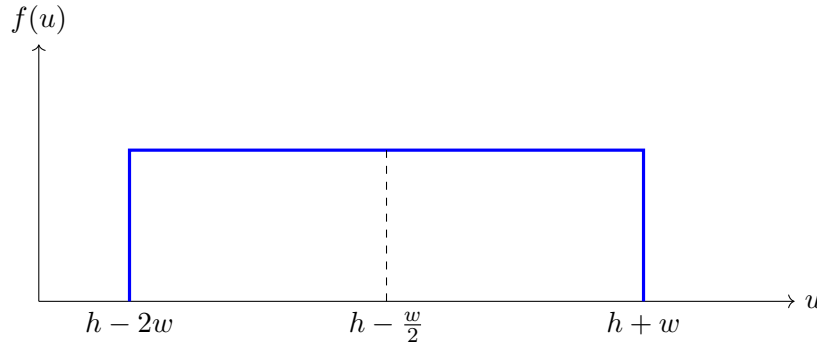
$$\sigma_U = \sqrt{\frac{3w^2}{4}} = \frac{\sqrt{3}}{2} w \quad (w > 0).$$

The center $h - \frac{w}{2}$ depends on both h and w , while the width $b - a = 3w$ depends only on w . Case A $\rightarrow$ B keeps w fixed, so width and spread stay the same, and the model shifts right by 5. Case C $\rightarrow$ D keeps h fixed and increases w ,

so width grows from 12 to 21, spread increases, and the center moves from 28 to 26.5 (left). If h and w both increase by 2, then width increases by $3(2) = 6$ (widens) and

$$E[U]' - (E[U]) = \left(h + 2 - \frac{w + 2}{2}\right) - \left(h - \frac{w}{2}\right) = 1,$$

so it also shifts right by 1.



15 Triangular with Parameters

Problem. A machine-cycle time T (seconds) is modeled by

$$T \sim \text{Triangular}(k, k + r, k + 3r), \quad r > 0.$$

Questions/Tasks.

- Derive $E[T]$, $\text{Var}(T)$, and σ_T in terms of k and r .
- State the validity conditions on k and r , and explain why they are sufficient.
- How do k and r separately affect center and spread?
- Compare shape with the symmetric triangular model having the same endpoints k and $k + 3r$. Is this model symmetric, left-skewed, or right-skewed?
- For fixed k , compare $r = 2$ and $r = 5$: discuss changes in mean, variance, and skewness direction.

C4

For $\text{Triangular}(a, c, b)$:

$$E[T] = \frac{a + b + c}{3}, \quad \text{Var}(T) = \frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}.$$

With $a = k$, $c = k + r$, $b = k + 3r$:

$$E[T] = \frac{k + (k + r) + (k + 3r)}{3} = k + \frac{4r}{3}.$$

For variance,

$$\text{Var}(T) = \frac{k^2 + (k + 3r)^2 + (k + r)^2 - k(k + 3r) - k(k + r) - (k + 3r)(k + r)}{18} = \frac{7r^2}{18}.$$

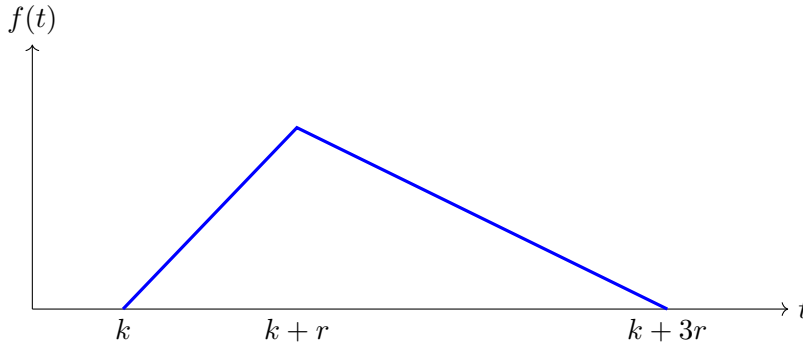
Thus

$$\sigma_T = \sqrt{\frac{7r^2}{18}} = r\sqrt{\frac{7}{18}}.$$

Validity requires $a \leq c \leq b$, i.e.

$$k \leq k + r \leq k + 3r,$$

which holds whenever $r \geq 0$; with $r > 0$, spread is nonzero, so the model is valid and non-degenerate for any real k . Parameter k shifts location only. Parameter r increases both mean offset ($\frac{4r}{3}$) and spread ($\propto r^2$). For endpoints k and $k + 3r$, symmetry would need mode at midpoint $k + \frac{3r}{2}$, but here mode is $k + r$, left of midpoint; therefore the model is right-skewed. Comparing $r = 2$ and $r = 5$ (fixed k): mean increases by $\frac{4}{3}(5 - 2) = 4$, variance scales from $\frac{28}{18}$ to $\frac{175}{18}$, and skewness direction stays right-skewed.



16 Normal with Parameters

Problem. The daily amount N (GB) of network data used by a class is modeled by

$$N \sim N(50 + 3p, (2 + q)^2), \quad p \in \mathbb{R}, \quad q > -2.$$

Questions/Tasks.

- Derive $E[N]$, $\text{Var}(N)$, and σ_N in terms of p and q .
- Explain the role of p and q : which controls location and which controls spread?
- Compare Case A: $(p, q) = (0, 1)$ and Case B: $(p, q) = (-2, 1)$. What changes in center, concentration around the mean, and tail likelihood?
- Compare Case C: $(p, q) = (0, 1)$ and Case D: $(p, q) = (0, 4)$. What changes in center, spread, concentration near the mean, and likelihood of values far from the mean?
- Between models $M_1 : (p, q) = (1, 0)$ and $M_2 : (p, q) = (-1, 3)$, which model is more likely to produce values at least 15 GB above its own mean? Justify conceptually.

C4

For $N(\mu, \sigma^2)$, $E[N] = \mu$, $\text{Var}(N) = \sigma^2$, $\sigma > 0$. So

$$E[N] = 50 + 3p, \quad \text{Var}(N) = (2 + q)^2, \quad \sigma_N = 2 + q \quad (q > -2).$$

Parameter p changes only location (horizontal shift). Parameter q changes spread: larger q gives larger σ , wider curve, lower peak, and heavier tails. Case A $\rightarrow$ B keeps $q = 1$, so $\sigma = 3$ is unchanged: concentration and tail behavior relative to each mean stay the same, but mean shifts from 50 to 44 (left by 6). Case C $\rightarrow$ D keeps $p = 0$, so mean stays 50, but σ increases from 3 to 6: values are less concentrated near 50 and far-from-mean values become more likely. For M_1 , $\sigma = 2$; for M_2 , $\sigma = 5$. A value 15 GB above its own mean is $15/\sigma$ standard deviations above mean: 7.5σ for M_1 versus 3σ for M_2 . The 3σ event is much less extreme, so M_2 is more likely.

